Synoptic CB: Porewater Nutrients

September 2025 Samples

2026-08-21

Contents

##Run Information

```r
cat("Run Information: NAME ") #lets you know what section you're in
```

```
## Run Information: NAME
```

```r
#set the run date & user name
  run_date <- "06/30/2026"
  sample_year <- "2025"
  sample_month <- "SEPTEMBER"
  user <- "Stephanie Wilson"

##Data entered incorrectly
Old_ID_1 <- c("MSM_202509_UP_LysC_45cm ")
New_ID_1 <- c("MSM_202509_UP_LysC_45cm")

#identify the files you want to read in
  #read in as a list to accommodate multiple runs in a month
  NOx_files <- c("Raw Data/SEAL_COMPASS_Synoptic_NOx_202509_1.csv",
                 "Raw Data/SEAL_COMPASS_Synoptic_NOx_202509_2.csv",
                 "Raw Data/SEAL_COMPASS_Synoptic_NOx_202509_3.csv")
  NH3_PO4_files <- c("Raw Data/SEAL_COMPASS_Synoptic_NH3_PO4_202509_1.csv",
                     "Raw Data/SEAL_COMPASS_Synoptic_NH3_PO4_202509_2.csv",
                     "Raw Data/SEAL_COMPASS_Synoptic_NH3_PO4_202509_3.csv",
                     "Raw Data/SEAL_COMPASS_Synoptic_PO4_202509_3_GOOD.csv")

# Define the file path for QAQC log file - NO Need to change just check year
  file_path <- "Raw Data/SEAL_COMPASS_Synoptic_QAQC_Log_2025.csv"
  final_path <- "Processed Data/COMPASS_Synoptic_Nutrients_202509.csv"

#record any notes about the run or anything other info here:
  run_notes <- "On the last NH3/PO4 run the PO4 standard curve was bad and it seemed like reagents were

#Set up file path for metadata
  #downloaded metadata csv - downloaded from Google drive as csv for this year
  Raw_Metadata = "Raw Data/COMPASS_SynopticCB_PW_SampleLog_2025.csv"
```

##Setup

##Read in metadata and create similar sample IDs for matching to samples

0.1 Import Data & Clean

##Fix data entered incorrectly

```
####Fix Sample IDs####
df_all$Sample_Name[df_all$Sample_Name == Old_ID_1] <- New_ID_1


#move the date to the correct place in the ones entered incorrectly
df_all$Sample_Name <- sub(
  "^(GCW|GWI|MSM|SWH)_(TR|WC|UP|SW)_([0-9]{6})_(LysA|LysB|LysC)_(10|20|45|NA)cm$",
  "\\1_\\3_\\2_\\4_\\5cm",
  df_all$Sample_Name
)


df_all$Sample_Name <- sub(
  "^(GCW|GWI|MSM|SWH)_(SW)_([0-9]{6})_(A|B|C)$",
  "\\1_\\3_\\2_\\4",
  df_all$Sample_Name
)
```

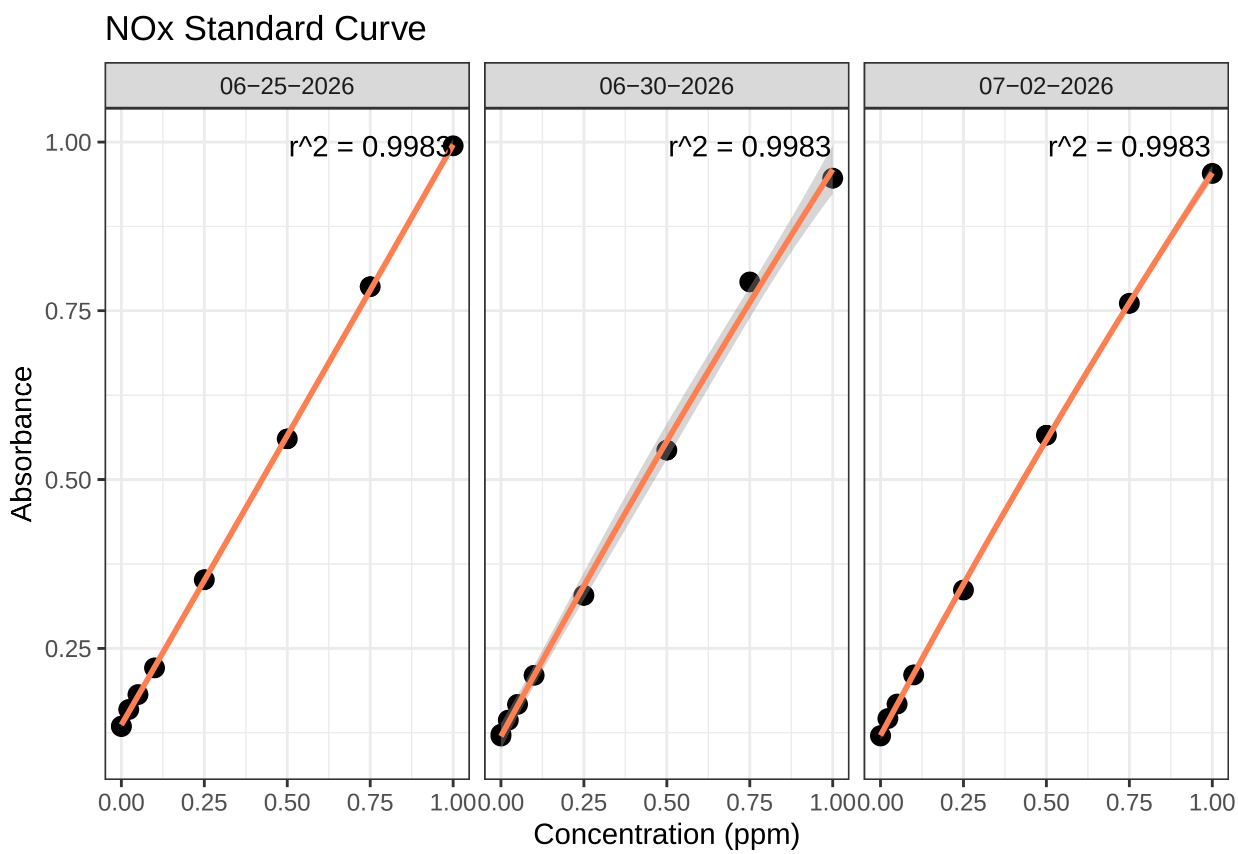
0.2 Assessing standard Curves

#Pull out standards data

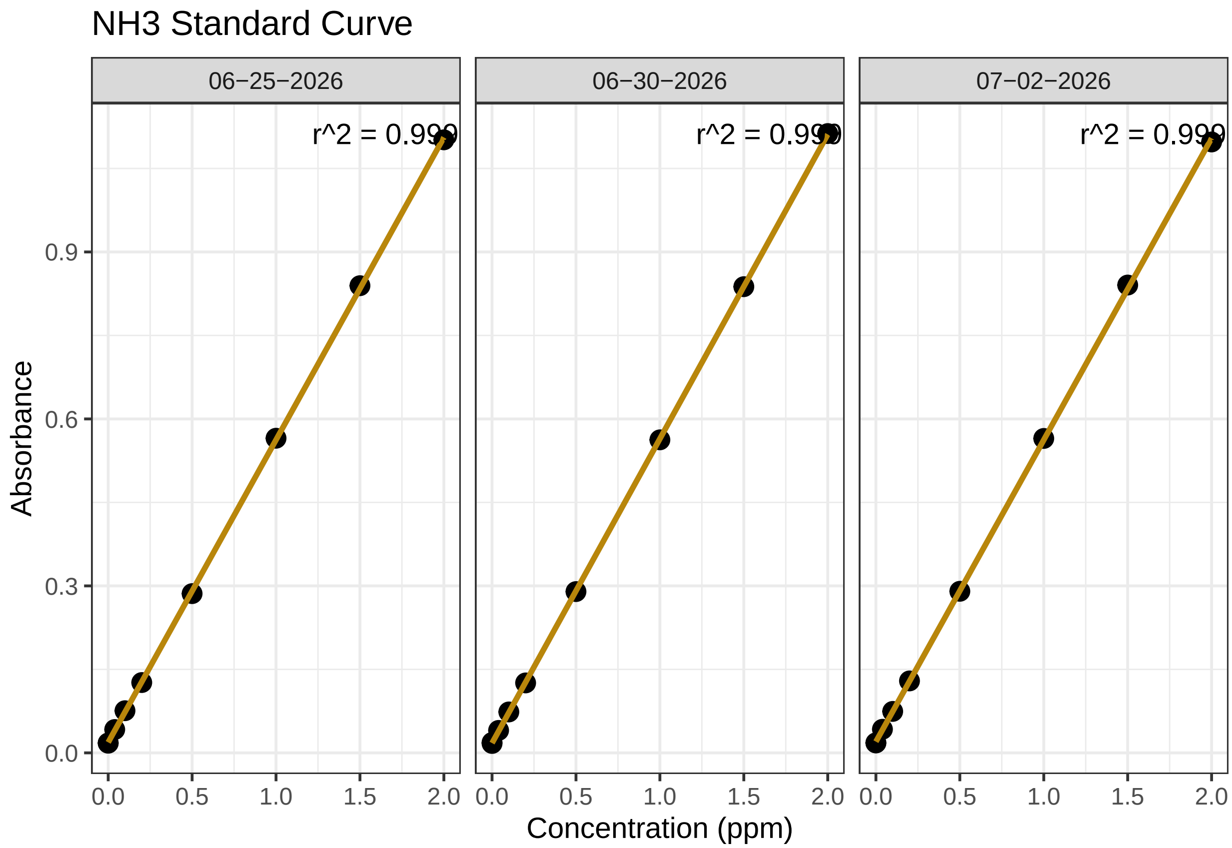
```
## Assess Standard Curves
```

0.3 Plot standards data

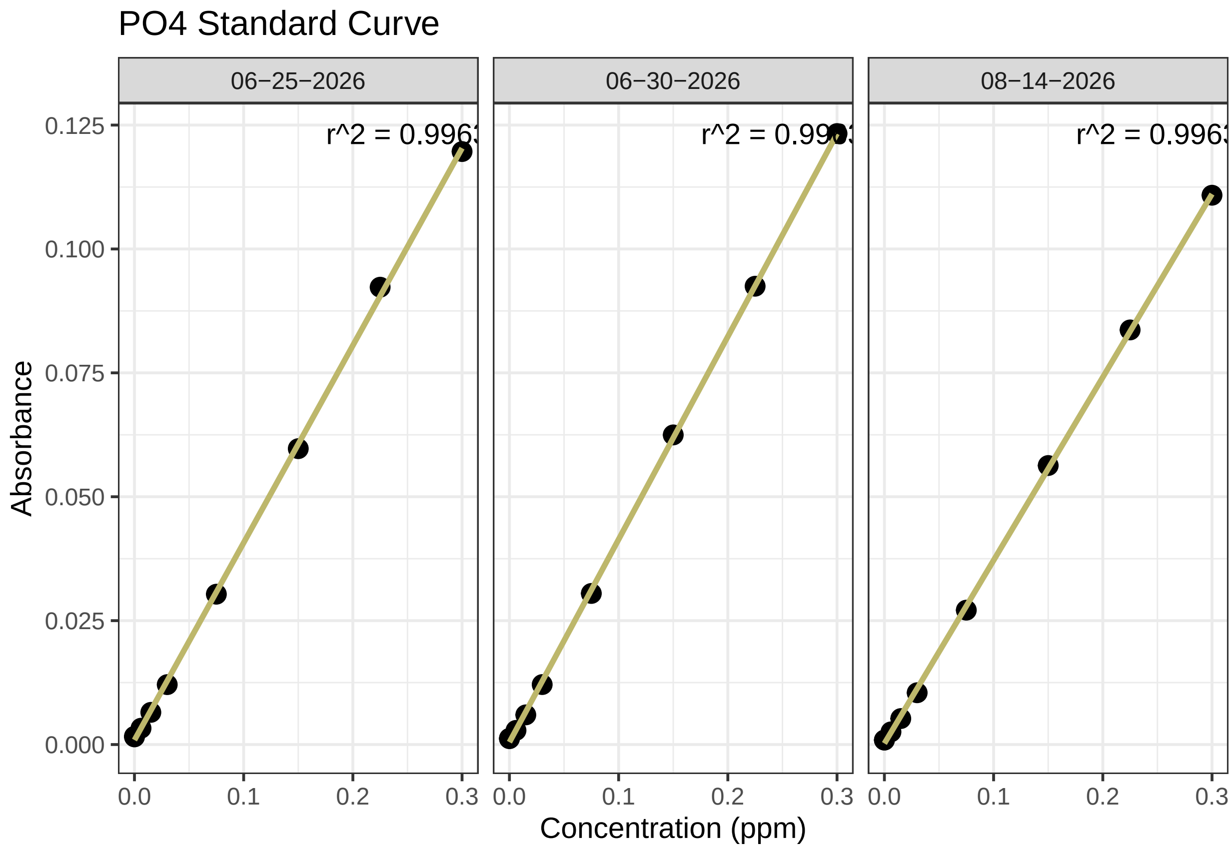
```
## Assess Standard Curves
```




```
## `geom_smooth()` using formula = 'y ~ x'
```




```
## `geom_smooth()` using formula = 'y ~ x'
```




```
## [1] "NOx Curve r2 GOOD - PROCEED"

## [1] "NH3 Curve r2 GOOD - PROCEED"

## [1] "PO4 Curve r2 GOOD - PROCEED"

## [1] "QAQC log file exists and has been read into the code."
```

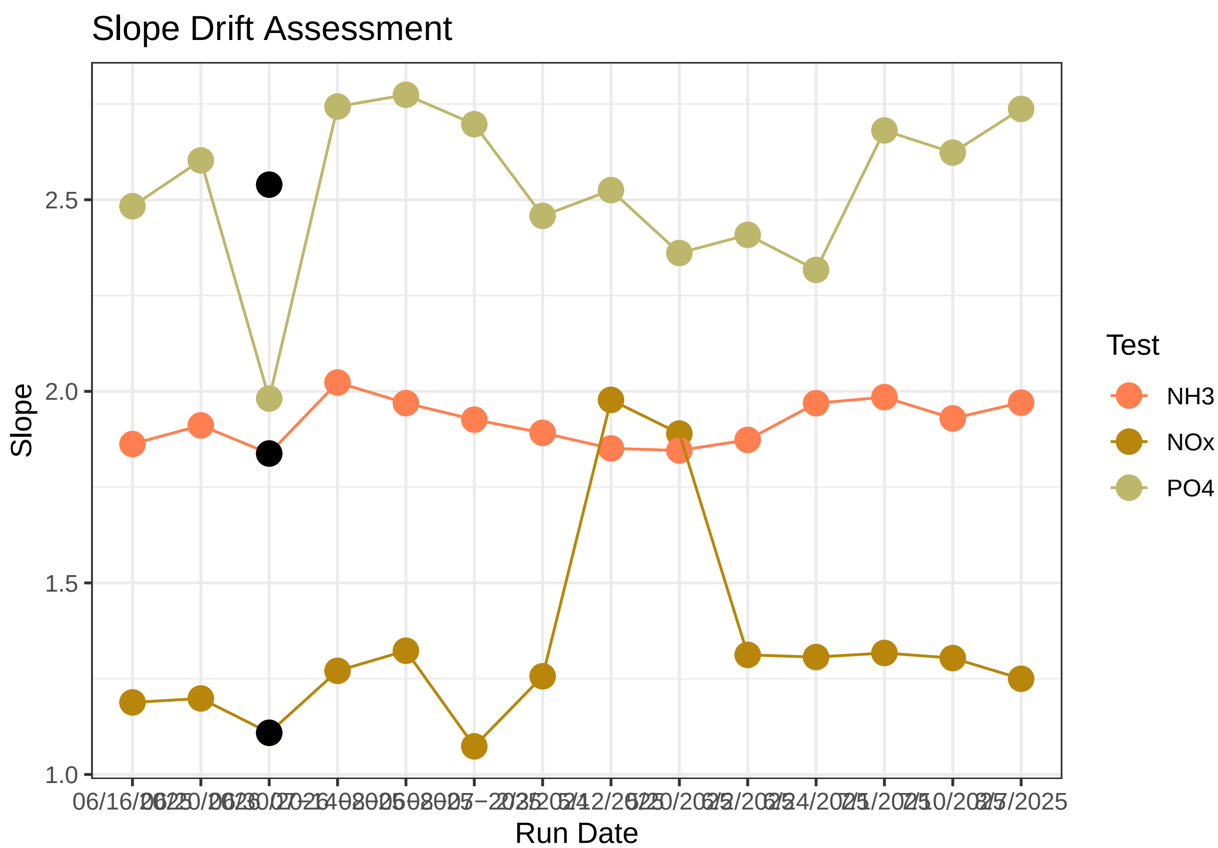


Table 1: Average Slope by Analyte

Test	avg_slope
NH3	1.917
NOx	1.341
PO4	2.528

0.4 Dilution Corrections - ensure the latest dilution is kept

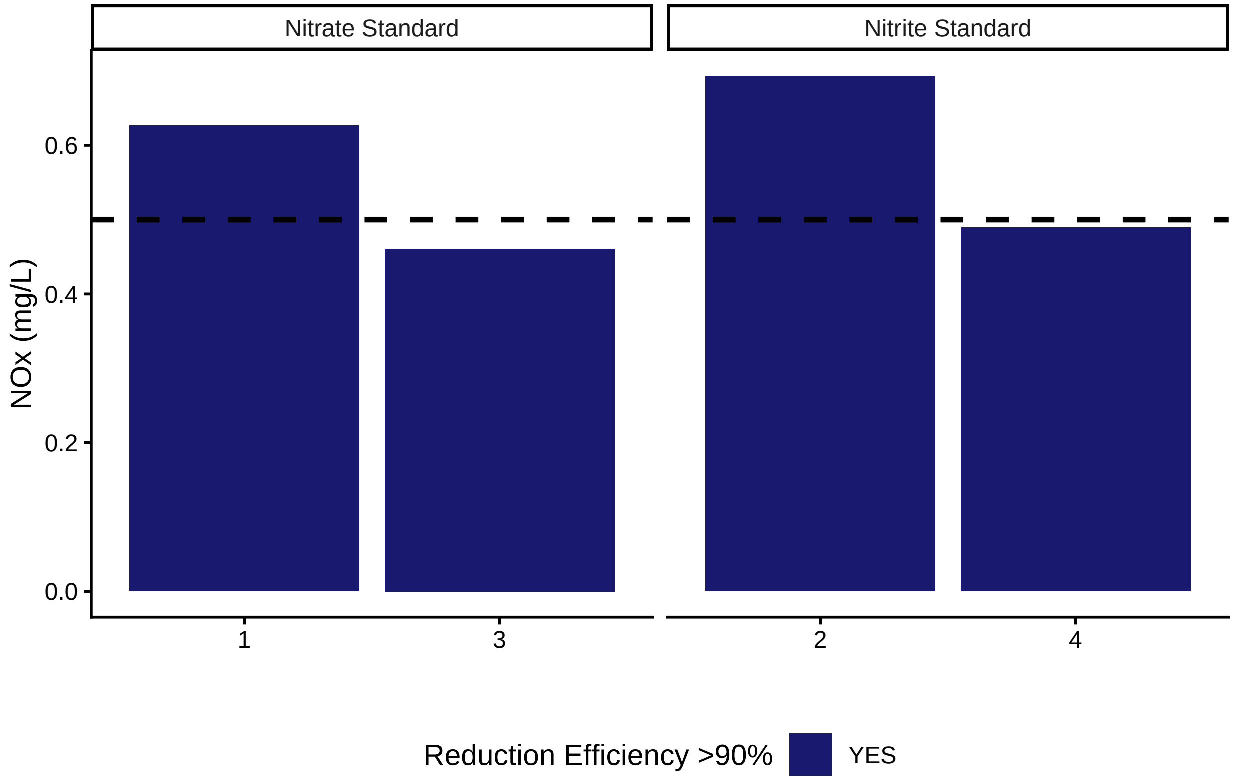
```
## Dilution Corrections

## Duplicated samples: GCW_202509_WC_LysA_45cm, GCW_202509_WC_LysB_10cm, GCW_202509_WC_LysB_20cm, GCW_2

##
##  All duplicated samples have valid dilutions. No naming issues detected.
```

0.5 Performance Check

```
## [1] "NOx pe Check has a % Difference <25% - PROCEED"

## Run mean = 1.550669

## Expected  = 1.51

## [1] "NH3 pe Check has a % Difference <25% - PROCEED"

## Run mean = 1.097768

## Expected  = 1.034

## [1] "PO4 pe Check has a % Difference <25% - PROCEED"

## Run mean = 0.8710093

## Expected  = 0.824
```

##Check NOx Reduction Efficiency

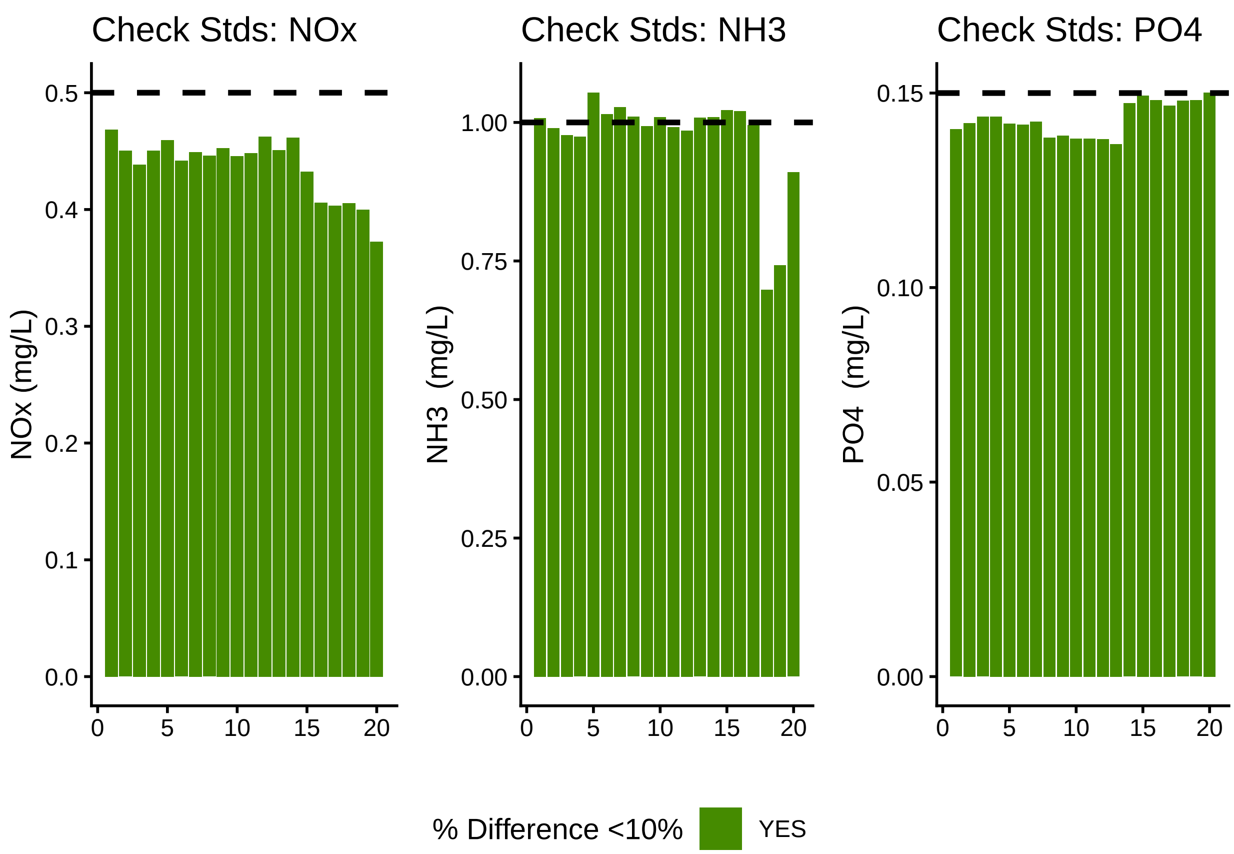
```
## Assess Reduction Efficiency
```




```
## [1] "Mean NOx Reduction Efficiency >95% - PROCEED"

## [1] 113.4839
```

0.6 Analyze the Check Standards

```
## Analyze Check Standards

## [1] "NOx Check Standard RSD within Range - PROCEED"

## [1] "NH3 Check Standard RSD within Range - PROCEED"

## [1] "PO4 Check Standard RSD within Range - PROCEED"
```




```
## [1] ">60% of NOx Check Standards are within range of expected concentration - PROCEED"

## [1] ">60% of NH3 Check Standards are within range of expected concentration - PROCEED"

## [1] ">60% of PO4 Check Standards are within range of expected concentration - PROCEED"
```

0.7 Analyze Blanks

```
## Assess Blanks

## [1] ">60% of NOx Blanks are below the lower 25% quartile of samples or 1/2 detection limit - PROCEED

## [1] ">60% of NH3 Blanks are below the lower 25% quartile of samples - PROCEED"

## [1] ">60% of PO4 Blanks  are below the lower 25% quartile of samples- PROCEED"
```

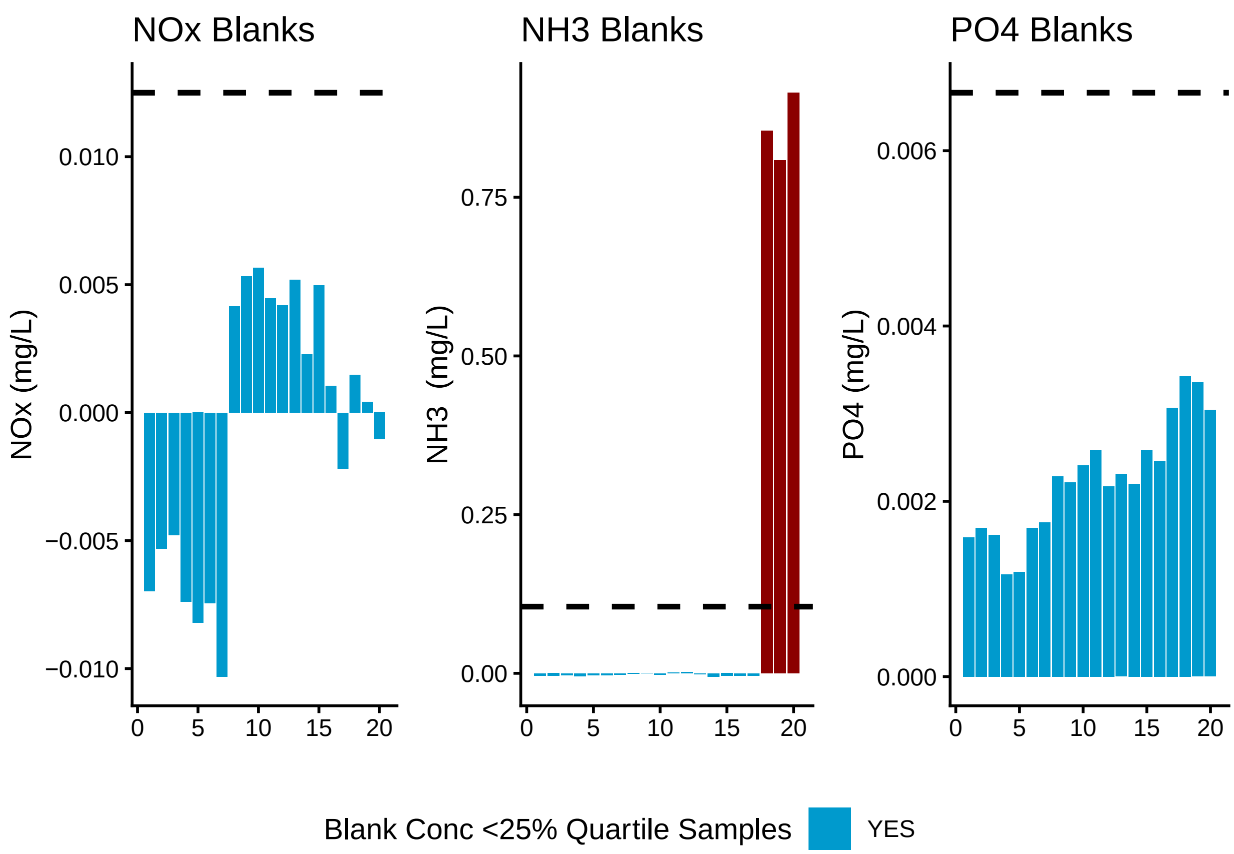
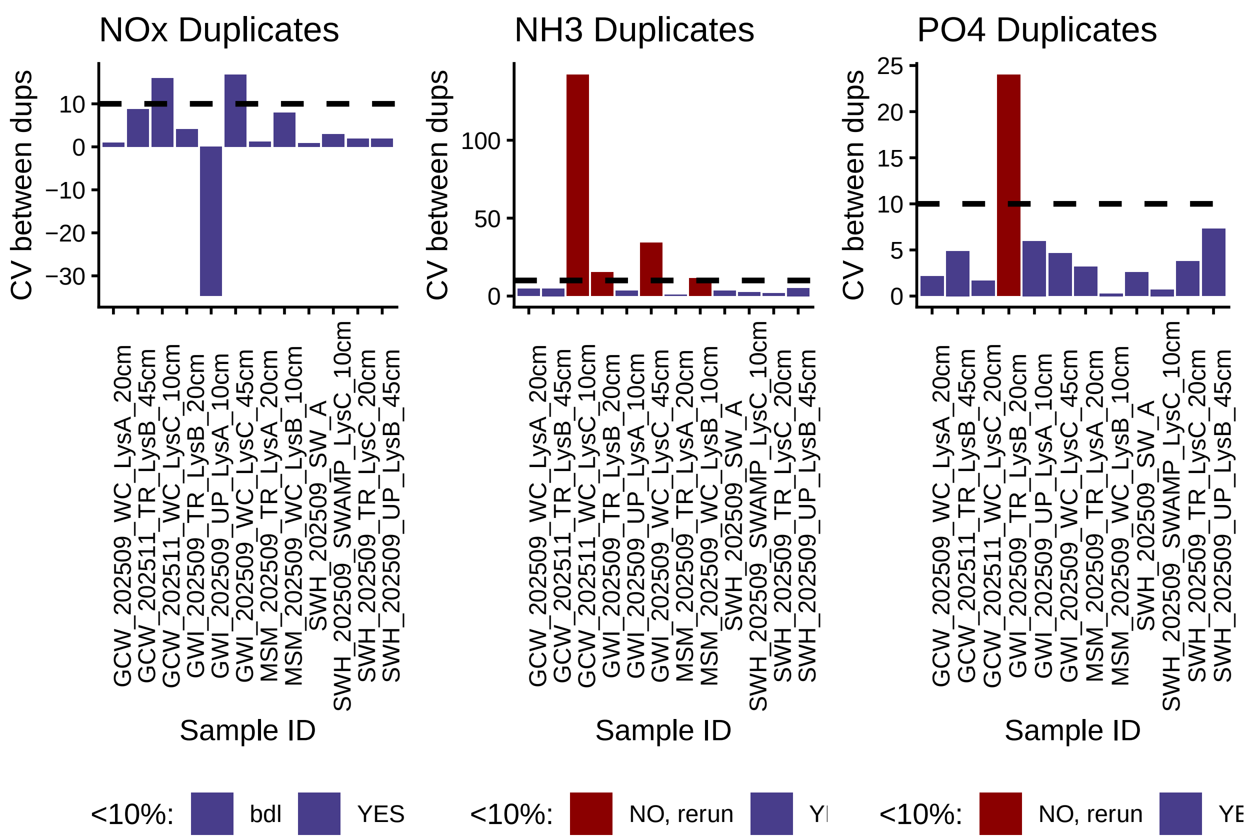


Table 2: Mean Concentration of Blanks

Test	Blank_Mean_Conc
NOx	-0.0007
NH3	0.1269
PO4	0.0022

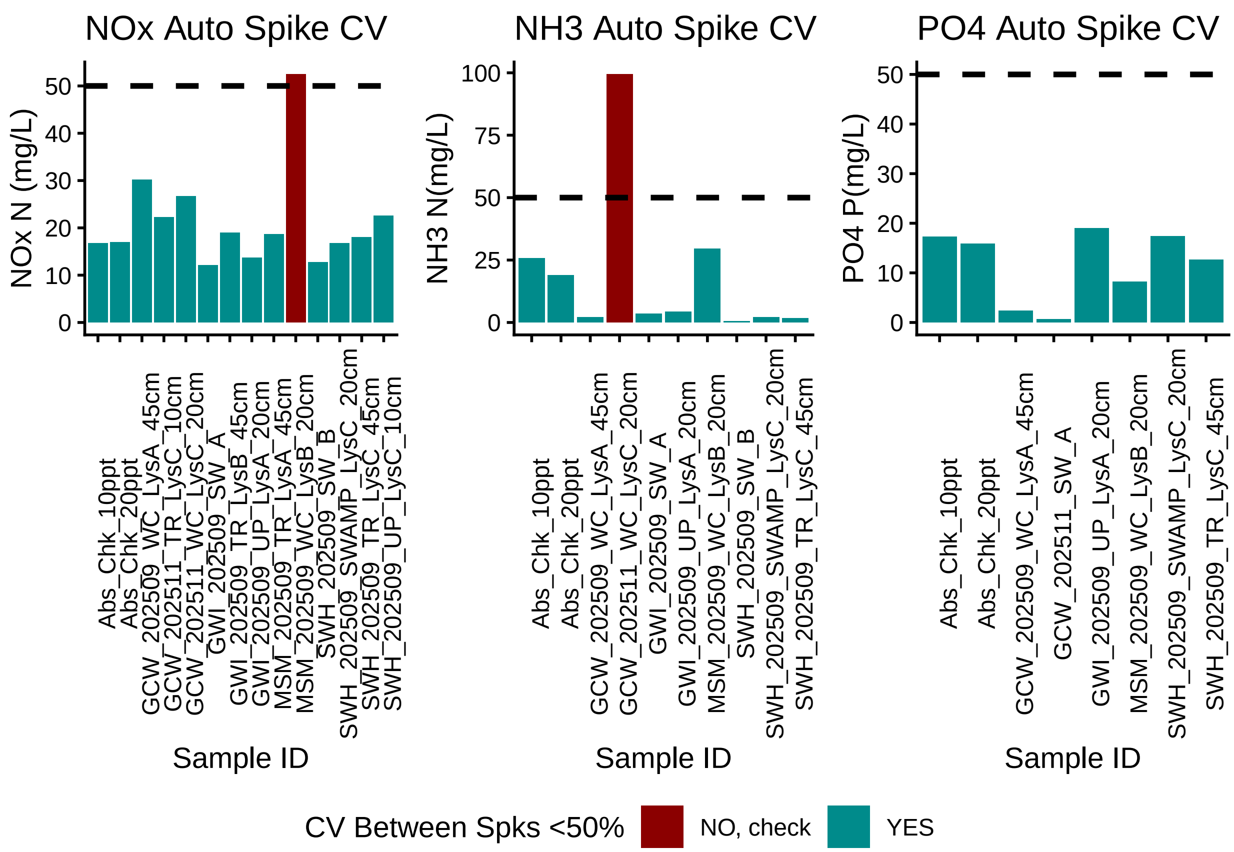
0.8 Analyze Duplicates

```
## Analyze Duplicates

## [1] ">60% of NOx Duplicates have a CV <10% - PROCEED"

## [1] ">60% of NH3 Duplicates have a CV <10% - PROCEED"

## [1] ">60% of PO4 Duplicates have a CV <10% - PROCEED"
```



0.9 Spikes

```
## [1] ">60% of Spikes have a CV <50% - PROCEED"

## [1] ">60% of Spikes have a CV <50% - PROCEED"

## [1] ">60% of Spikes have a CV <50% - PROCEED"

## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



0.10 Matrix Effects

```
## [1] "NO NOx Matrix Effect, PROCEED"

## [1] "NO NH3 Matrix Effect, PROCEED"

## [1] "NO PO4 Matrix Effect, PROCEED"
```

0.11 Unit Converted Data Column Added (mg/L to uM)

0.12 Sample Flagging - Within range of standard curve

```
## Sample Flagging
```

0.13 Pull out sample id information

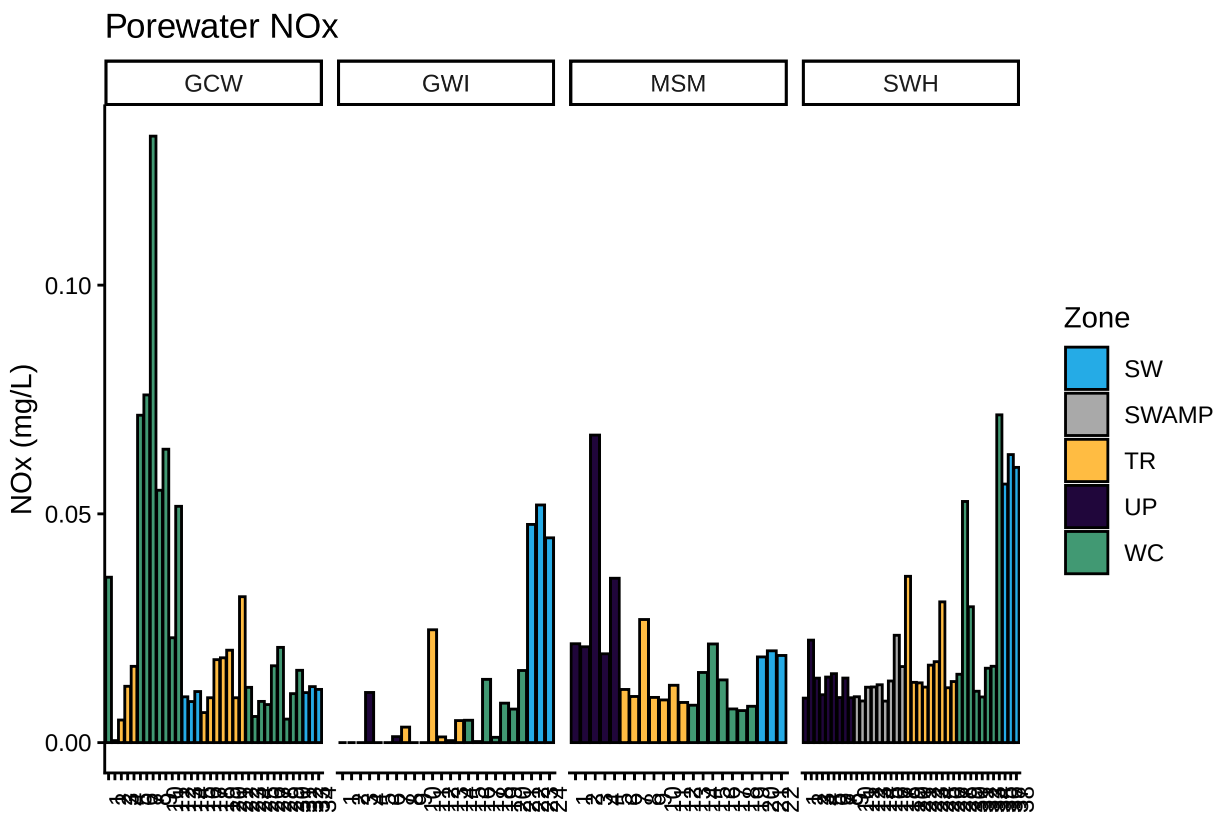
```
## Sample Processing
```

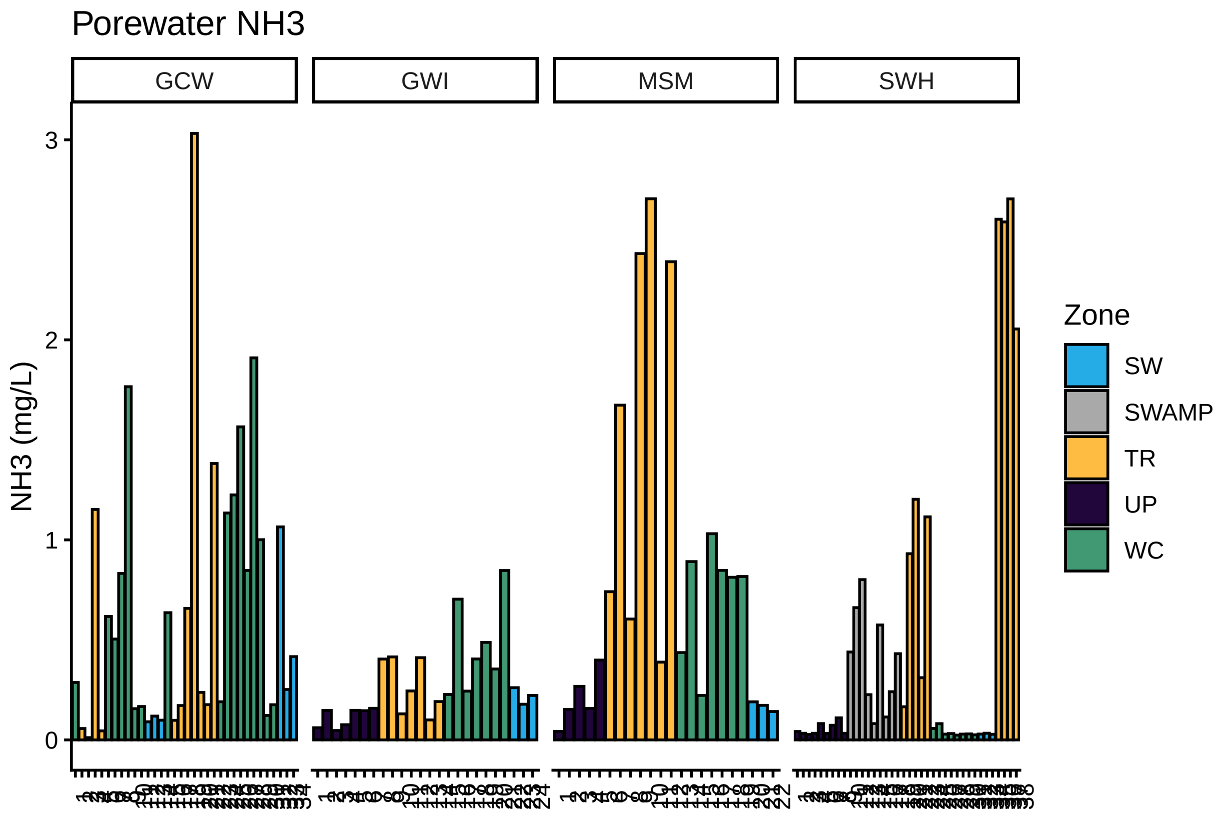
0.14 Check to see if samples run match metadata & merge info

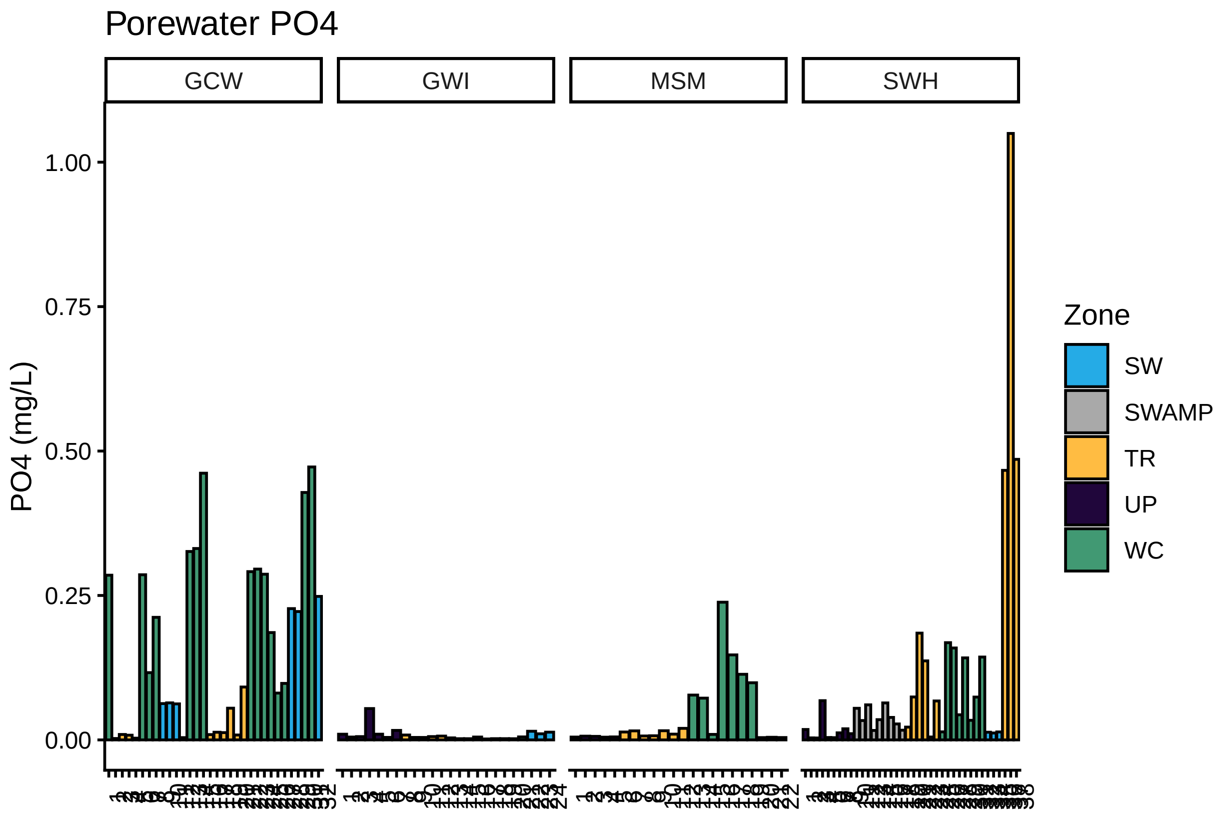
```
## Check Sample IDs with Metadata

## All sample IDs are present in metadata.
```

0.15 Visualize Data

```
## Visualize Data
```




Porewater NH3
GCW
GWI
MSM
SWH
NH3 (mg/L)
0
1
2
3
Zone
SW
SWAMP
TR
UP
WC



0.16 Export Processed Data

#end